

Neal Kaushik Sharma

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EDUCATION

Iowa State University

B.S. in Computer Science Honors, Minors in Artificial Intelligence, Cybersecurity, and Data Science

GPA: 3.84/4.0

Ames, Iowa

Expected: May 2028

Junior

COMPUTATIONAL SKILLS

Languages: Java, PHP, React, Python, C, C++, C#, SQL, HTML, CSS, TypeScript, JavaScript

Tools and Frameworks: Git, Spring Boot, Flask, Django, .NET, Laravel, Postman, SSH, Azure, LaTeX

EXPERIENCE

Application Development Intern – Fareway Stores

June 2026 - August 2026

- Developed and deployed the Planogram Tracking Dashboard, a full-stack internal application used across 144 Fareway stores by supervisors and leadership to track planogram execution, monitor store compliance, and analyze thousands of monthly store responses.
- Built the application using Laravel/PHP, React, PostgreSQL, Redis, and external APIs, following Fareway codebase practices while designing database structures, backend services, frontend components, sync workflows, role-based access systems, and automated tests.
- Owned project planning through discovery, sprint planning, user stories, and demos while collaborating with the Web Development team.
- Contributed to additional modules through debugging, test cases, Bash scripting, and production maintenance.

Software Engineer Intern – Tata Consultancy Services

June 2025 - August 2025

- Worked as a Web Application Developer, where my main project was Brewmaster (simulated coffee store), a full-stack online retail system built from scratch using C#, ASP.NET Core MVC, SQL Server, ADO.NET, jQuery, and AJAX ([GitHub](#)).
- Developed core system features, implementing secure authentication, image uploads, password hashing, role-based access, centralized logging, and a full MVC architecture connected to real-time SQL Server to handle inventory, orders, and product management efficiently.
- Rotated with frontend, backend, database, DevOps, QA, and cloud engineers, gaining exposure to the software development lifecycle and Agile workflows while learning practical skills in API development, Bootstrap, Azure cloud, documentation, and presentations.

Cybersecurity & Software Development Assistant – ISEAGE

January 2026 - Present

- Support the planning and execution of five annual Cyber Defense Challenges (CDCs) by building and maintaining multi-application scenarios that simulate realistic operations, including web applications, services, and infrastructure, with hundreds of participants each competition.
- Manage Linux-based virtual machines hosting web applications, databases, and services; write scripts to deploy, start, stop, and reset services, configure networking between VMs, monitor resources, automate scenario setup, and integrate applications across multiple VMs.
- Debug software, fix broken services, and monitor systems during live competitions, responding to failures or attacks to keep scenarios running while teams defend against the penetration attempts of cybersecurity professionals.

Research - Malicious Unlearning Attacks against Predictive Uncertainty in AI Models

August 2025 - May 2026

- Research on how machine unlearning impacts AI model reliability, presented at the *National Conference on Undergraduate Research (NCUR)*.
- Implemented and compared unlearning methods (data removal from pre-trained models) on image classification models using medical and benchmark datasets, focusing on model behavior, calibration, confidence, accuracy, training time, and impact on predictive performance.
- Applied uncertainty estimation techniques (Deep Ensembles, Monte Carlo Dropout) and metrics such as Expected Calibration Error and Brier Score to quantify prediction confidence and identify vulnerabilities introduced by unlearning.

PROJECTS

Cy Saint's Hospital - Hospital Management Platform [[Android GitHub](#)] [[WebApp GitHub](#)]

August 2025 - May 2026

Built across COMS 3090 (Android) and COM S 3190 (React/Node.js) to streamline workflows for patients, doctors, pharmacists, and admins.

- COM S 3090 (Android): Backend developer implementing authentication, role-based access, appointments, prescription management, Firebase notifications, and AI health assistant features. Awarded *Best Project Overall (1st place)* among 300+ students.
- COM S 3190 (Web): Developed both frontend and backend for a React/Node.js full-stack application, building patient dashboards, doctor portals, pharmacy workflows, admin management, JWT authentication, MongoDB APIs, notifications, and health record features.

Space-Miner - 2D Java Game [[GitHub](#)]

October 2025 - Present

Personal project: A top-down survival/adventure game built entirely from scratch in Java, under active development with plans for public release.

- Developed core gameplay systems including player movement, collision detection, mining mechanics, particle effects, loot drops, companion AI using A* search, inventory, crafting, dynamic lighting, day/night cycles, procedural tile-based worlds, and screen culling.
- Built supporting engine features including minimap, UI systems, save/load functionality, sound integration, and in-game debugging tools.

View additional projects: nealks.netlify.app/projects